Cloud Computing Enablers:-

Virtualization:-

- Creates Virtual resources.
- Uses a Hypervisor.
- Creates Virtual Machines.

Web Services:-

- Connect applications through the internet.
- Use APIs for Communication.
- SOAP is a web service Protocol.
- REST is a Simple web service style.

Cloud Deployment:-

- Easy resources allocation
- Fast application access.
- Resources Sharing.

Cloud Services Models:-

Iaas:-

-> Provides infrastructure.

-> Provider Manages hardware and Network.

-> User Manages OS, applications and data.

Paas:-

-> Provides a development Platform

-> Provides Manages infrastructure and Platform.

-> users Manages application and data.

Saas:-

-> Provides ready-to-use Software.

-> Provider Manages almost-Everything.

-> User Mainly uses the Software.

Definition:-

→ Computing resources through the internet.

Characteristics:-

→ on-demand access - use resources when needed.

→ Resource Pooling - Resources are shared.

→ Elasticity - Resources increases (or) decreases automatically.

→ Scalability - handles More (or) Less workload.

→ Measured Service - usage is Moritored

Benefits:-

→ Low cost

→ Easy access

→ Flexible resources.

→ Easy resource allocation
→ Fast application access
→ Resource sharing

↓

Scalability

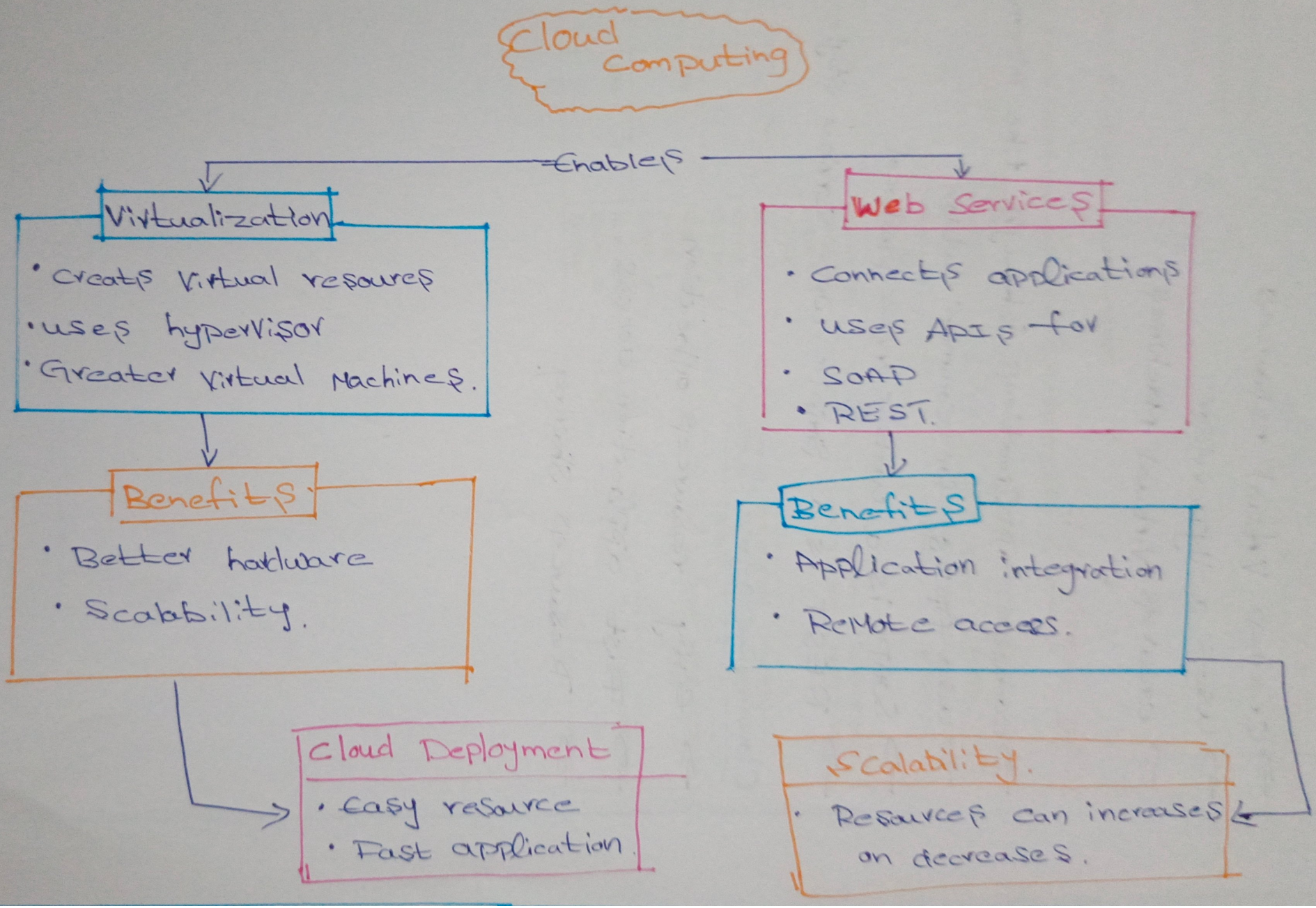
→ Resources can increase or decrease based on demand.

- Infrastructure layer, platform layer, application layer, shared responsibility

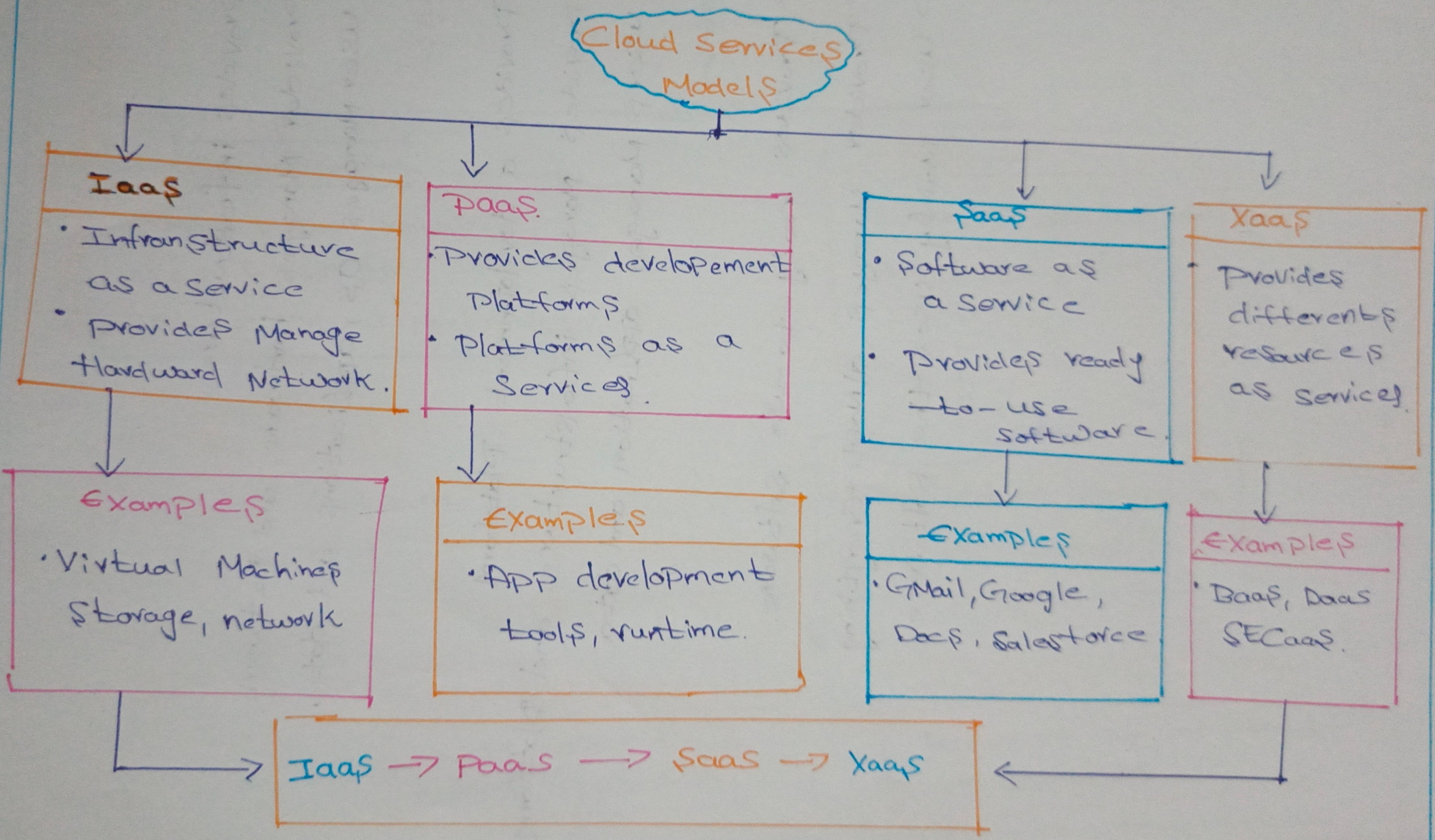
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

4. Concept Map: Virtualization and Web Services :-



3. Concept Map: Service Models.



Hardware Evolution:-

→ Mainframe Computers
→ Personal Computers
→ Servers
→ Powerful Hardware.

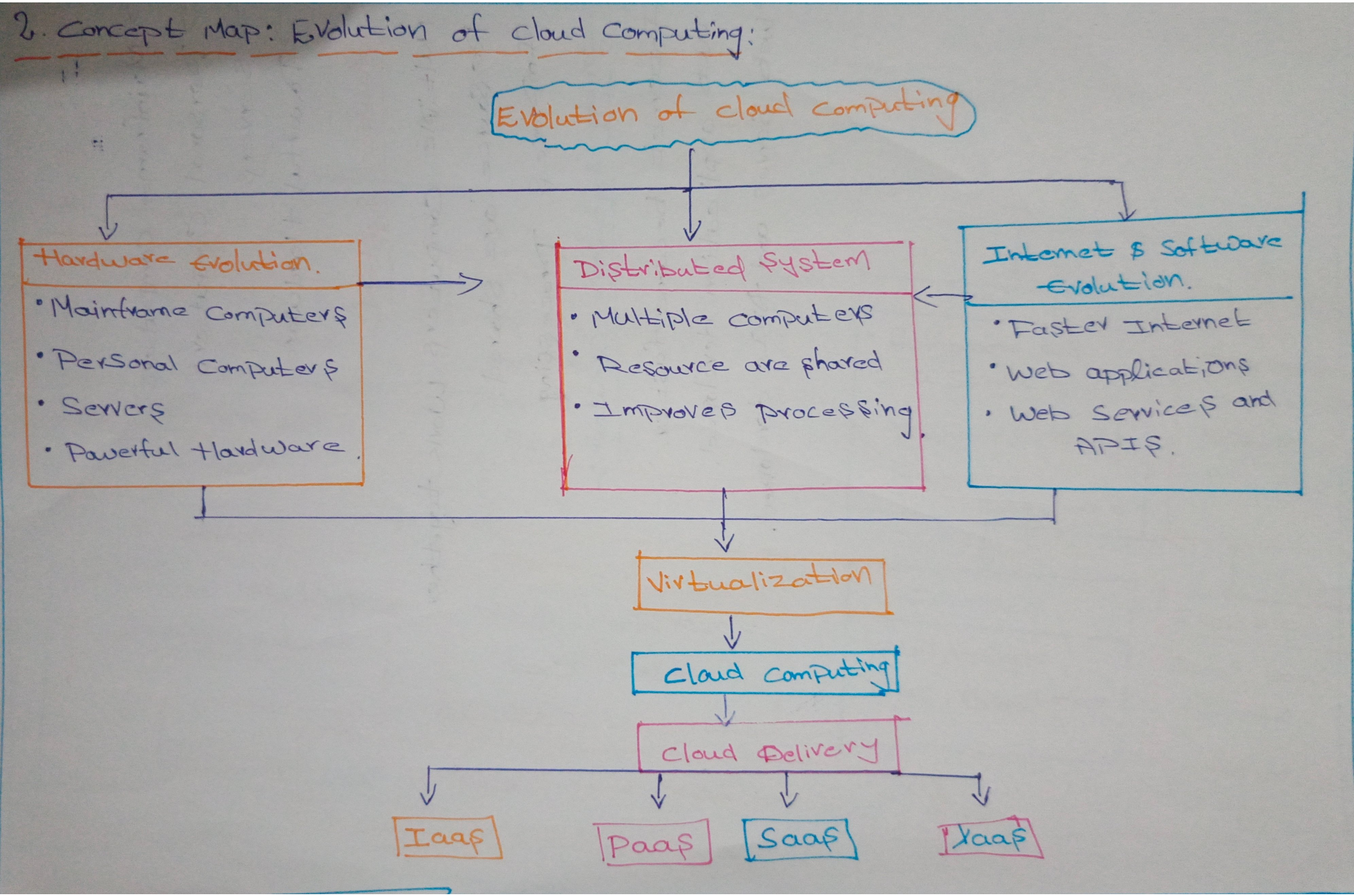
Distributed System:-

→ Multiple Computers work together
→ Resource are Shared.
→ Improves Processing.

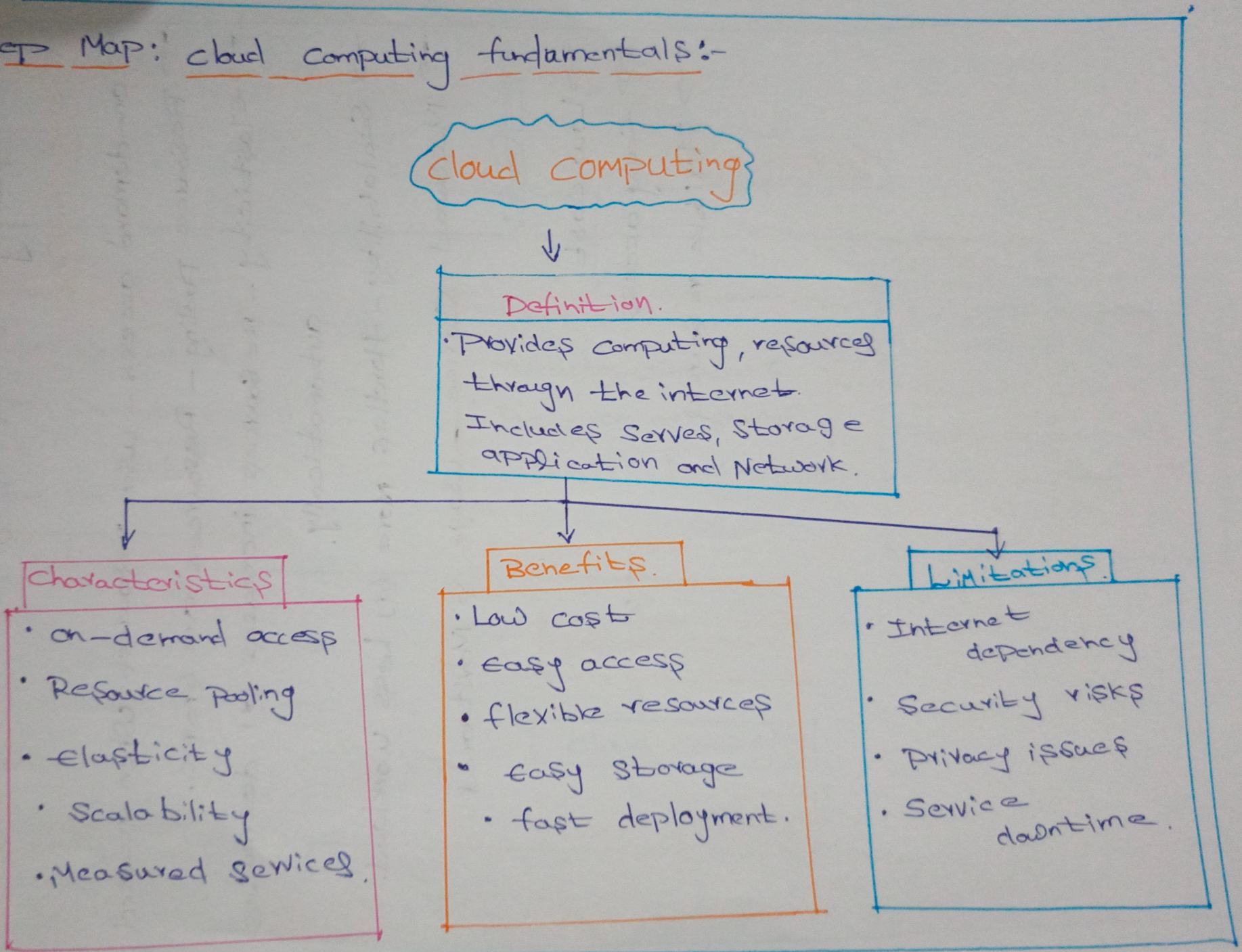
Internet Software Evolution:-

→ Internet became faster.
→ Web application developed.
→ Web Servers and APIs developed.

2. Concept Map: Evolution of cloud computing:



1- Concept Map: cloud computing fundamentals:-



SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**

Question Count: 4

Code of Conduct

II A.Tharun kumar reddy(192572068) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.tharun Kumar Reddy

Assessment Tasks Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer